ENTERPRISE PLATFORM MODERNIZATION

Salesforce to Microsoft Power Platform Migration

Role: Senior Business Analyst & Project Lead **Domain:** Life Sciences / Pharmaceuticals

Executive Summary

When a global pharmaceutical leader engaged our team to migrate 20+ business-critical applications off Salesforce onto Microsoft Power Platform, it seemed straightforward: replicate everything, preserve every screen, and do it in nine months.

It was neither straightforward nor nine months.

As the Senior BA and Project Lead managing a globally distributed team across the US, Europe, and Asia-Pacific, I recognised what I came to call the "Replica Fallacy" — the dangerous assumption that a low-code platform built for agility could simply absorb years of layered Apex triggers, hidden validation logic, and 500k+ record data operations without fundamental re-architecture. Surface-level UI screen counts had masked the true complexity beneath.

Rather than continue reverse-engineering spaghetti logic, I pivoted the entire programme from imitation to intent. Through a formal Logic Inventory protocol and asynchronous stakeholder workshops, I reframed every requirement around a single question: *what is this process actually trying to achieve?* That shift alone eliminated 40% of cosmetic bugs and unlocked a more performant Model-Driven App architecture that the original Canvas-first approach could never have delivered.

The outcome: 18 months, double the initial effort estimate, not the numbers in the original slide deck — but the right numbers, earned through technical integrity rather than wishful planning. Custom-engineered solutions replaced expensive third-party middleware including a third-party ETL middleware integration handling 500k+ weekly records, and the consolidated Dataverse environment generated a material reduction in annual platform licensing costs.

1. The Challenge: Navigating the "Replica Fallacy"

The project was initiated with a rigid stakeholder mandate: a 1:1 functional replica ('same screens, same speed') on a low-code architecture that was never designed for high-code Salesforce parity.

- **The Logic Trap:** Initial estimations relied on surface-level UI screen counts, missing layered Apex triggers and hidden validation rules.
- **Architecture Gaps:** High-volume data loads (500k+ records) that were instantaneous in Salesforce faced timeouts and delegation limits in Power Apps.
- **Complexity Bloat:** Many legacy apps were unnecessarily complicated, requiring intensive reverse engineering of "spaghetti logic" rather than modern requirement gathering.

2. Strategic Intervention & Remote Leadership

Faced with a timeline that threatened to double, I pivoted the project from 'Reverse Engineering' to Intent-Based Discovery.

- **Global Stakeholder Management:** Managed alignment across US, Europe, and Asia-Pacific using asynchronous workshops (Jira) to maintain momentum across time zones.
- **The Logic Audit:** Implemented a formal "Logic Inventory" protocol to simplify legacy processes before estimation, replacing "T-shirt sizing" with technical reality.
- **Framework Pivot:** Negotiated the shift from UI-heavy Canvas Apps to performance-optimized Model-Driven Apps (MDA) for complex relational entities, leveraging native responsiveness to eliminate 40% of cosmetic bugs.

3. Technical Innovation & Value Delivered

To bridge significant platform gaps, I oversaw the engineering of high-impact custom utilities.

- **Global Search Connector:** Overcame native Canvas limitations by engineering a custom connector for seamless cross-entity search.
- **Scalable Data Flows:** Replaced expensive Boomi integrations with native Dataflows to support weekly loads of 500k+ records without API throttling.
- **Advanced Multi-Attachment:** Developed a custom asynchronous workflow combining JavaScript and Power Automate to enable robust multi-attachment support in Dataverse.
- **Localization**: Engineered a localization utility for full Japanese language support within Canvas Apps

4. Project Performance & ROI

Metric	Initial Estimate	Final Outcome
Timeline	9 Months	18 Months
Effort / Cost	1.0x	~2.0x
Value Realization	• Significant Reduction in Licensing Cost (~10-20% of SFDC) • 5000+ Licenses Migrated to PowerApps • 300+ Licenses saved by identifying and de-scoping unutilized apps • Lower operational cost	

5. Key Lessons & Playbook for Future Migrations

- **Discovery ≠ Demo:** My biggest lesson was prioritizing business outcomes ("What is the intent?") over historical behavior ("How did the old button work?").
- **Load Test Early:** Production-level data volumes (500k+ records) must be tested within the first month to identify delegation bottlenecks.
- **Governance First:** Establish a Change Control Board (CCB) early to sign off on behavioral shifts between Salesforce and Power Platform.

Technical Stack: Power Apps (Canvas/Model-Driven), Dataverse, Power Automate, Dataflows, JavaScript, SQL, Salesforce (Apex/SOQL).